

SSC - 2 (1-4)

Analysis of Categorical Data

* Attribute

A Positive and negative association

* Yule's coefficient

$$Q = \frac{(AB)(\alpha\beta) - (\alpha B)(A\beta)}{(AB)(\alpha\beta) + (\alpha B)(A\beta)}$$

gurukul

	A	α
B	(AB)	(αB)
β	$(A\beta)$	$(\alpha\beta)$

Note:

* Value of Q lies between -1 and $+1$

* If $Q = +ve$ then two attributes are +vely associated
and If $Q = -ve$ then two attributes are -vely associated

* Yule's coefficient of colligation.

$$Y = \frac{1 - \sqrt{\frac{(AB) \times (\alpha\beta)}{(A\beta) \times (\alpha B)}}}{1 + \sqrt{\frac{(AB) \times (\alpha\beta)}{(A\beta) \times (\alpha B)}}}$$

	A	α
B	(AB)	(αB)
β	(A β)	($\alpha\beta$)

Note:

* If $Y = +ve$ then there is +ve association.

* If $Y = -ve$ then there is -ve association.

* Pearson's coefficient of Contingency

$$C = \sqrt{\frac{\chi^2}{N + \chi^2}}$$

$$\chi^2 = \text{chi-square}$$

where,

$$\chi^2 = \frac{N[(AB)(\alpha\beta) - (A\beta)(\alpha B)]^2}{(A)(\alpha)(B)(\beta)}$$

	A	α	Total
B	(AB)	(αB)	(β)
β	(A β)	($\alpha\beta$)	(β)
Total	(A)	(α)	(N)